Vaibhavi Kariya

Undergraduate Researcher, Deep Learning for Medical Image Analysis

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SUMMARY

Undergraduate researcher in Computer Engineering at CHARUSAT, working on deep learning for medical image analysis. First author of three research works spanning histopathology, dermoscopic skin lesion analysis, and brain MRI classification, including one accepted and presented paper and two journal manuscripts. Research interests include medical image analysis, image classification, and image segmentation.

SELECTED RESEARCH & PUBLICATIONS

HGC-Net (First Author) · Histopathology **ACCEPTED & PRESENTED · DOI AWAITED**

HGC-Net: Hybrid CNN-GraphSAGE-CBAM Framework for Atypical Mitosis Classification in Histopathology Images

ICST 2026, IIT Patna (1–2 Aug 2026). Co-author: Prof. Ronak R. Patel.

- Combined a ResNet-50 encoder, a three-layer GraphSAGE network over a cosine-similarity kNN patch graph and CBAM-1D channel attention; addressed 5.8:1 class imbalance with minority-class augmentation and Focal Loss.
- MIDOG 2025 test set: 94.89% accuracy, 90.29% recall, AUC-ROC 0.982.

CMQ-Net (First Author) · Dermoscopy **UNDER REVISION**

CMQ-Net: A Hybrid CNN–MedMamba Network with Quantum-Inspired Channel Attention and Adaptive Feature Fusion for Dermoscopic Skin Lesion Classification

BMC Medical Imaging (Q1, JIF: 3.9). Corresponding author: Prof. Ronak R. Patel.

- Built a three-stream model (CNN backbone, MedMamba state-space block, quantum-inspired channel attention) with gated feature fusion; analysed with ablations, Grad-CAM and Integrated Gradients.
- HAM10000 (7-class, balanced split): 98.34% accuracy, 98.31% F1, 99.92% macro-AUC, 1.86M parameters.

QGAN-DINOv2 (First Author) · Brain MRI **SUBMITTED**

QGAN-DINOv2: A Quantum-Augmented Vision Transformer Framework for Brain Tumor MRI Classification

Signal, Image and Video Processing (Q3, JIF: 2.7). Corresponding author: Prof. Ronak R. Patel.

- Designed a quantum-classical GAN for class-preserving augmentation and adapted a frozen DINOv2-Base backbone with LoRA (rank 8) on query/value projections.
- Fused CLS tokens from four transformer depths by learnable cross-attention: 95.19% accuracy, 95.10% macro-F1 on a four-class MRI benchmark.

RESEARCH EXPERIENCE

Summer Research Internship, CSPIT, CHARUSAT **11 May – 25 Jun 2026**

- Conducted research on atypical mitosis classification in histopathology images and prepared the project report and presentation under the supervision of Prof. Ronak R. Patel.

EDUCATION

B.Tech in Computer Engineering **2024 – 2028**

Charotar University of Science & Technology (CHARUSAT), Anand, Gujarat, India · CGPA 7.96 / 10

SKILLS

Research: Deep Learning, Medical Image Analysis, Computer Vision, Graph Neural Networks, ViT, LoRA, State-Space Models, Feature Fusion, Model Evaluation, Explainability

Programming: Python, C++, SQL

Libraries: PyTorch, TensorFlow/Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, NLTK, Flask

Tools: Git, GitHub

TECHNICAL EXPERIENCE

AI & Backend Developer Intern, LaunchED Global (Remote) **Dec 2025 – Feb 2026**

- Built a Python/Flask AI study assistant with machine-learning topic clustering and NLP-based text summarisation.

CERTIFICATIONS & ACHIEVEMENTS

- **ICST 2026, IIT Patna:** Certificate of Presentation (HGC-Net) Aug 2026
- **NPTEL (IIT Madras):** Design and Analysis of Algorithms (66%); Theory of Computation (77%) 2026
- **IBM / Coursera:** Data Analysis with Python; IBM AI Fundamentals (Language and Vision in AI) 2026
- **Additional:** Power BI Data Analyst Certificate; IBM Qiskit Fall Fest; GirlScript Summer of Code 2026